Marcus **Sjolin**

Product and Technical Leader

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Summary of Skills

- Python, C++ and React
- Robot Operating System
- Databricks and Apache Spark
- Vim, Git

Work Experience

NVIDIA — *Santa Clara, CA, USA*

Manager, Technical Product Management - Autonomous Vehicles — *October 2023 - Present*

- Defined roadmap, priorities and KPIs for Active Safety, Level 2, Level 2+ and Level 3 autonomy for Mercedes-Benz and Jaguar Land Rover
- Delivered NVIDIA's first production Level 2 AV and Active Safety systems for global deployment in next-gen Mercedes-Benz vehicles
- Drove cross-functional execution and risk management for 100+ engineers in Perception, Localization, Planning and Controls
- Owned feature compliance with EU General Safety Regulations to enable vehicle homologation and market certification
- Achieved highest possible 5-star safety ratings in Euro NCAP, NCAP AD and equivalent global programs

Embark Trucks — *San Francisco, CA, USA*

Director, Technical Product Management and Systems Engineering — *September 2020 - August 2023*

- Founded and scaled the Product organization to ~25 people (reporting to the CTO), managing both ICs and managers across Product Management, Data Science, Systems Engineering, Release, Simulation Ops, and Triage
- Defined the technical roadmap with C-suite and Engineering leads, aligning work of 120+ engineers across software and hardware
- Led a 20+ person team that developed autonomy capabilities in downtown Oakland, directly contributing to Embark's $295M SPAC merger
- Wrote product requirements to deploy mature trucks into Knight-Swift's fleet as part of the Truck Transfer Program
- Developed company-level North Star metrics and analyses that improved reliability and performance by orders of magnitude
- Established structured requirements management, traceability and V&V planning across autonomy stack and supporting services

Product Lead — *August 2019 - September 2020*

- Led special project of 15+ engineers that developed and hardened autonomous features for $70M Series C fundraise led by Tiger Global
- Managed a team of engineers and data scientists and wrote code and requirements as an individual contributor
- Filed patent for a state-machine-based pipeline to automatically extract on-road scenarios using Apache Spark
- Prototyped and analyzed safety metrics like TTC, PET and more in Databricks based on literature like Mobileye's Responsibility-Sensitive Safety
- Built pipeline to extract on-road issues and automatically upload to Jira for prioritization and triage

Software Engineer — *May 2017 - August 2019*

- Led team of 10+ that built the first truck to drive 300+ miles autonomously from Phoenix to LA, resulting in $30M Series B led by Sequoia
- Developed real-time lane tracking using imperfect lane and object data in C++ to optimize functional and compute performance
- Built an on-truck diagnostics system to record compute resource utilization across ROS nodes and wrote data pipelines to analyze results

Meta (WhatsApp) — *Menlo Park, CA, USA*

Software Engineer, Full Stack — *January 2017 - April 2017*

- Implemented client-side emoji suggestion logic in React for WhatsApp Web, streamlining search and improving input efficiency

Wish — *San Francisco, CA, USA*

Software Engineer, Full Stack — *April 2016 - September 2016*

- Re-architected Android network and image libraries across all Wish apps, contributing to a 10–14% increase in average session length

Hipmunk — *San Francisco, CA, USA*

Software Engineer, Android & iOS — *September 2015 - December 2015*

- Modernized mobile UX and implemented new travel planning features across Android and iOS as part of the Hipmunk Discover rollout

Pebble Technology — *Palo Alto, CA, USA*

Software Engineer, Android — *January 2015 - May 2015*

- Developed Android integration for Calendar and Weather features on the Pebble Time smartwatch, leveraging The Weather Channel API

Projects & Distinctions

Drivable Surface Detection — *ECE 457B (Waterloo, Canada)*

Python, TensorFlow, NumPy — *April 2018*

- Implemented a SegNet (using VGG) deep convolutional neural network to detect drivable surface using a fusion of camera and LiDAR data

Berkeley Artificial Intelligence Course — *edX (Online)*

Python – Constraint Propagation, Markov Decision Processes etc. — *January 2015 - May 2015*

Stanford Machine Learning Course — *Coursera (Online)*

Python, Octave – Logistic Regression, SVMs, Clustering etc. — *June 2014 - September 2014*

Education

University of Waterloo — *Waterloo, ON, Canada*

Bachelor of Software Engineering, Honors, Co-op — *2018*